Evaluating the Robustness of Lightweight Machine Learning Models to Label Noise

Garima
Dept. of Computer Science & Engineering
Chitkara University
Punjab, India
garima313.be22@chitkara.edu.in

Gurnoor Dhaliwal
Dept. of Computer Science & Engineering
Chitkara University
Punjab, India
gurnoor344.be22@chitkara.edu.in

Gagan Deep
Dept. of Computer Science & Engineering
Chitkara University
Punjab, India
gagandeep310.be22@chitkara.edu.in

Divesh Gupta
Dept. of Computer Science & Engineering
Chitkara University
Punjab, India
divesh291.be22@chitkara.edu.in

Dr. Shikha Tuteja*
Associate Professor
Dept. of Computer Science & Engineering
Chitkara University
Punjab, India
shikha.1290@chitkara.edu.in

***Abstract*—Label noise is a common issue in real-world datasets and can significantly degrade the performance of models. This study examines *performance degradation contrast to asymmetric noise* the robustness of six classical machine learning models—Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and LightGBM—under two types of label noise: symmetric and asymmetric. Experiments are conducted on two datasets, namely Breast Cancer Wisconsin and Adult Income, to evaluate the generalizability of model behavior. We add noise only to the training data at different levels (0%, 10%, 20%, and 30%), while evaluation is performed on clean test data with Accuracy and F1-score metrics.**

Results show that symmetric noise leads to significantly higher performance degradation as compared to asymmetric noise across all models. Out of the tested models, SVM demonstrates strong and consistent robustness across most settings, while Decision Tree exhibits the least robustness to label corruption. Additionally, F1-score is observed to be more sensitive to noise in the data, while accuracy does not change as much—especially when one class has significantly more samples than the other. This highlights the importance of selecting noise-robust models and suitable evaluation metrics in real-world scenarios where data quality cannot be guaranteed.

Keywords - Label noise, machine learning robustness, symmetric noise, asymmetric noise, SVM, LightGBM, Decision Tree, noisy labels, classification.

I. Introduction

Machine learning models depend on the assumption that training data is accurately labeled. However, in real-world scenarios, this assumption is often violated due to errors in data labeling, ambiguous data instances, automated labeling pipelines, or mismatches in the data collection processes. Such inaccurate or corrupted labels are commonly referred to as label noise [8], and they can weaken the model's performance and reliability.

Label noise is especially important because it directly affects the learning signal during training. Unlike feature noise, which corrupts the input attributes, label noise corrupts the target signal itself [10]. For this reason, models may pick up wrong decision boundaries, which leads to poor performance and unstable predictions This problem becomes even more important in domains such as healthcare, finance, and decision-support systems, where bad predictions can have serious consequences.

To better understand this problem, label noise is often classified into two major types: *symmetric noise* and *asymmetric noise*. In symmetric noise, class labels are randomly flipped with a certain probability, representing uniform corruption across all classes. In comparison, asymmetric noise introduces structured or directional corruption, where labels are changed from one class to another in a planned manner. These two noise settings allow researchers to simulate both random and realistic label corruption scenarios.

While a large body of research has centered on deep learning approaches for dealing with noisy [11], [12], older machine learning models are still widely used, especially for tabular datasets. Classical models such as Logistic Regression, Decision Trees [17], Random Forests [4], Support Vector Machines (SVM) [15], K-Nearest Neighbours (KNN), and gradient boosting models, for example LightGBM [7] widely used when working with tables of data. However, their robustness under controlled label noise conditions is not always well understood, particularly in a comparative and multi-dataset setting.

In this work, we have studied the robustness of six classical machine learning models by adding variable amount of symmetric and asymmetric label noise. For this experiment, we have used two different standard benchmark datasets: One is a smaller, clean and numerical dataset i.e. Breast Cancer Wisconsin Dataset, and the other is a larger heterogenous dataset with combined feature types i.e. Adult Income Dataset. This set-up enables us to study and evaluate whether robustness patterns are similar across different types of data.

In order to ensure a controlled experiment and fair evaluation, we have introduced noise only in the training data. However, only the clean, untouched data has been tested. For measuring performance, both Accuracy and F1-score are used. Other than these metrics, relative degradation and variance are used across various folds.

The main contributions of this work are as follows:

- Provides a systematic comparison of six standard machine learning models under specific types of noise.

- A cross-dataset evaluation that shows how robustness varies with different dataset characteristics.
- A detailed analysis of performance degradation using Accuracy and F1-score that clearly shows the differences in these metrics.
- Models that are consistently robust or highly sensitive to noise or discrepancies in data, are identified.

The results of this study demonstrate the importance of robustness analysis as real-world machine-learning applications often face unpredictable conditions. The study also provides useful guidance for choosing models when data is noisy.

II. LITERATURE REVIEW

In the world of supervised classification, label noise has long been considered a persistent challenge as corrupted labels are known to interfere with the learning signal. This can weaken the ability of model to generalise. Recent survey works show that noisy-label learning is still an active research area with major approaches including noise-robust learning, data-cleansing, sample selection and loss correction[9]. While most of this work focuses on deep learning, the above described challenge is not limited to it. With corrupted labels the decision boundaries become unstable and can lead to failure in reflecting true performance.

In noisy-label research, it is common to evaluate models under controlled synthetic corruption, as this allows performance degradation to be studied in a systematic manner [9], [10], [14]. The two standard settings used for such experiments are symmetric noise and asymmetric noise [9], [10], [14]. Symmetric noise refers to the random flipping of class labels, while asymmetric noise captures class-dependent or directional corruption. This distinction is important because the two settings influence classifiers in different ways: symmetric noise weakens the training signal more uniformly across classes, whereas asymmetric noise can resemble realistic annotation bias or one-sided confusion between specific categories [9], [10], [14].

Robustness has become a topic of significant interest among current studies on deep neural networks, especially those focused on computer vision and medical imaging tasks. Noise-tolerant learning techniques have gained prominence through approaches such as transition matrix estimation, sample selection, loss correction, data cleaning, and confident learning [9], [12], [13], [14]. On the other hand, there remain several tabular data applications that depend on classical machine learning models due to their simplicity in learning and interpretation and their compatibility with small or structured datasets. This trend can be used to conduct comparative research on the performance of classic classifiers when exposed to noise without noise mitigation techniques.

Benchmark dataset choice also matters in robustness research. Using only one dataset can make findings overly dataset-specific, so cross-dataset comparison is often necessary to show that the patterns seen are stable [9]. In this work, the Breast Cancer Wisconsin (Diagnostic) dataset and the Adult dataset provide two contrasting test beds. The breast cancer data from Wisconsin contains features computed from digitized images of fine needle aspirates of breast masses [1], while the Adult dataset is a larger tabular dataset with diverse feature categories and missing data [2]. Since these datasets have different size, feature structure, and domain, they are suitable for examining whether model robustness trends are similar across \different classification settings.

Therefore, rather than creating a new noise-correction method, the main aim of this work is to compare and analyse which conventional models remain consistent and which ones quickly collapse. This study also shows whether baseline accuracy truly reflects robustness when the training data is noisy. The focus is on keeping the work practical for tabular-data workflows where quality of label is not guaranteed.

III. METHODOLOGY

A. Research Objective

The objective of this study is to evaluate the robustness of conventional machine learning classifiers under controlled label noise. The idea is to see how different models perform when wrong labels are added to the training data, whereas testing data remains clean. Performance results on different datasets, noise types and amount of noise are compared using standard classification metrics.

B. Datasets

Two We have used two benchmark datasets with tabular classification. The first dataset - Breast Cancer Wisconsin dataset [1] is comparatively smaller, numerical and clean. This makes it suitable for planned testing and debugging of the pipeline. The second dataset - Adult Income dataset [2] is larger and more diverse. It offers a more realistic tabular setting with higher complexity. It was used to check whether the robustness patterns observed, remain consistent on a second dataset. Hence, the use of two datasets helps lower the risk of conclusions being specific to only one data distribution.

TABLE I Summary of Datasets Used in the Study

Dataset	Instances	Features	Task Type	Source
Breast Cancer Wisconsin	569	30	Binary Classification	UCI ML Reposito ry [1]
Adult Income	48,842	14	Binary Classification	UCI ML Reposito ry [2]

C. Experimental Models

Six classical supervised machine learning models that we have chosen for robustness analysis and comparison are: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine(SVM), K-Nearest Neighbours(KNN), and LightGBM. These specific models were selected because each model follows different learning approach - linear, tree-based, margin-based, instance based, ensemble and boosting.

TABLE II Classical Machine Learning Models Used in the Study

Model	Category	Working Principle	Reason
Logistic Regression	Linear	Linear boundary via probabilities	Baseline model
Decision Tree	Tree	Hierarchical rule-based splits	Studies split sensitivity
Random Forest	Ensemble	Averaged trees on random subsets	Improves robustness
SVM	Margin	Max-margin hyperplane	Strong tabular performance
KNN	Instance	Nearest neighbor voting	Sensitive to noisy labels
LightGBM	Boosting	Sequential boosted trees	Powerful tabular learner

D. Experimental Pipeline

The entire experimental workflow was set up as a controlled robustness pipeline. The following are the steps in pipeline:

Baseline Training: In case of each dataset, the original clean dataset was first used to establish a baseline performance. Data was split into training and testing sets, and their Accuracy and F1-score were recorded.

Cross-Validation Set-up: In order to make results more reliable and avoid dependency on a single unit , 5-fold stratified cross-validation [3] was applied. Stratification ensured that class proportions were preserved across folds.

Noise Injection: As already mentioned, label noise was added only to training dataset (at different levels: 0%, 10%, 20%, 30%), while test labels were kept untouched. This was done to ensure true generalisation. Symmetric noise randomly flipped a set percentage of labels, while asymmetric noise flipped labels in one direction from one class to another.

Model Training and Evaluation: For For each dataset, noise type, noise level, and fold, all the six models were trained on the noisy training set and tested on clean set.

TABLE III Noise Settings and Evaluation Criteria

Component	Setting / Value
Noise Types	Symmetric noise, Asymmetric noise
Noise Rates	0%, 10%, 20%, 30%
Noise Injection Target	Training labels only
Test Set	Clean and untouched
Validation Strategy	5-fold stratified cross-validation
Performance Metrics	Accuracy, F1-score
Additional Analysis	Relative degradation, standard deviation, robustness ranking

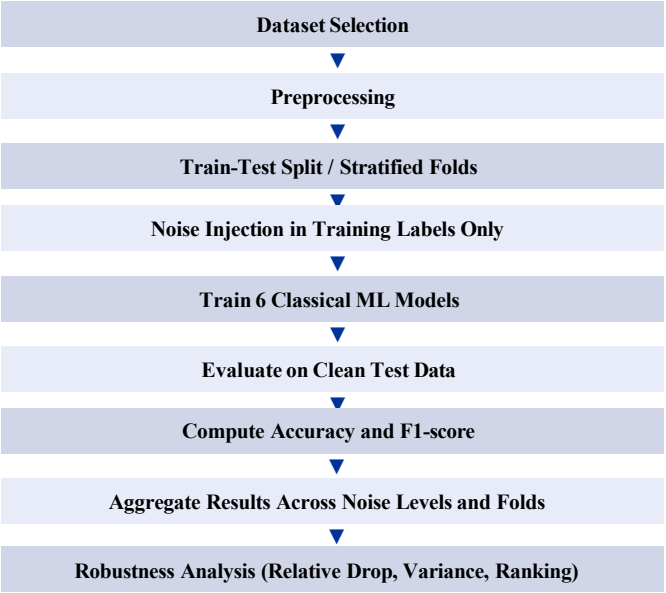

Fig. 1. Overall workflow of the proposed robustness evaluation framework for classical machine learning models under label noise.

E. Performance Measures

Accuracy: Accuracy shows the number of correct predictions out of all cases.

F1-score: F1-score was used as a balanced performance metric as it is a combination of precision and recall. It works well in situations where data is unevenly distributed across classes.

Relative Degradation: We have used this metric to quantify robustness and measure the drop in performance from clean baseline. It is measured as :

$$RPD(\eta) = (M(0) - M(\eta)) / M(0) \times 100\%$$

This shows how performance of each model declines as noise increases.

Standard Deviation Across Folds: The standard deviation of performance across cross-validation folds was used to analyze model stability.

F. Result Aggregation and Analysis

After the experiments were done, we combined fold-wise outputs in summary tables. Reported mean and standard deviation of Accuracy and F1-score, for each dataset, noise type and level, and model. Analysis was mainly on performance drop, degradation trends, robustness rank and symmetric versus asymmetric noise across models.

G. Reproducibility and Implementation

Ran all experiments in Python using a notebook-based pipeline. Kept separate notebooks/scripts for baseline code, noise generation, cross-validation and analysis. Also, used a fixed random seed to improve reproducibility of results.

IV. RESULTS AND DISCUSSION

This section focuses on the experimental findings from the Breast Cancer Winconsin and Adult Income datasets under controlled symmetric and asymmetric label noise. The discussion focuses not only on raw performance, but also on

robustness trends, degradation behavior, and differences between models across datasets. Since the objective of this work is to study robustness rather than only clean-data accuracy, the interpretation emphasizes how performance changes as noise increases.

A. Baseline Performance on Clean Data

Before introducing noise, all models were evaluated on clean training data to establish a reference point. On the Breast Cancer Wisconsin dataset, baseline performance was uniformly high, indicating that the dataset is relatively well-structured and easier to classify. SVM achieved the strongest baseline performance with a mean accuracy of 97.7% and F1-score of 98.2%, followed closely by Logistic Regression (97.4% / 97.9%) and LightGBM (96.7% / 97.4%). KNN and Random Forest also performed strongly, with accuracies of 96.3% and 95.4%, respectively, while Decision Tree showed the lowest baseline performance at 91.6% accuracy and 93.4% F1-score.

A different pattern emerged on the Adult Income dataset, where overall performance was lower because of greater data complexity, mixed feature types, and class imbalance. LightGBM achieved the best baseline accuracy of 87.4% with an F1-score of 71.3%, followed by Random Forest at 85.9% accuracy and 67.9% F1-score, and SVM at 84.9% accuracy and 63.5% F1-score. Logistic Regression and KNN showed moderate baseline performance, while Decision Tree again produced the weakest baseline results at 81.4% accuracy and 61.8% F1-score. These results suggest that ensemble- and boosting-based methods are better suited to the Adult dataset, whereas margin-based learning is especially effective on the smaller and cleaner Breast Cancer Wisconsin dataset.

Table IV. Baseline mean accuracy and F1-score at 0% label noise for all models on the Breast Cancer Wisconsin and Adult Income datasets. BC = Breast Cancer Wisconsin. Sym. = Symmetric, Asym. = Asymmetric

Model	BC Acc.	BC F1	Adult Acc.	Adult F1
SVM	97.7%	98.2%	84.9%	63.5%
Logistic Reg.	97.4%	97.9%	82.5%	55.1%
LightGBM	96.7%	97.4%	87.4%	71.3%
KNN	96.3%	97.1%	82.9%	62.1%
Random Forest	95.4%	96.4%	85.9%	67.9%
Decision Tree	91.6%	93.4%	81.4%	61.8%

B. Effect of Symmetric Label Noise

Symmetric noise reduced performance on both datasets, with degradation increasing as the noise rate rose from 0% to 30%. Although all models were affected, the rate of deterioration differed substantially across classifiers.

On the Breast Cancer Wisconsin dataset, SVM was the most robust model under symmetric noise, with a total relative accuracy drop of only 6.5% at 30% noise. Logistic Regression remained the second most stable with a 9.2% relative drop, followed closely by Random Forest at 9.4%. In contrast, Decision Tree exhibited the strongest degradation with a 26.3% relative drop, confirming that greedy split-based learning is highly sensitive to incorrect labels. LightGBM also degraded substantially at 19.8%, while KNN showed a 16.9% relative drop at the highest noise level.

On the Adult Income dataset, the ranking changed slightly, but the overall pattern remained similar. SVM again emerged as the most robust model under symmetric noise, with only a 1.1% relative accuracy drop at 30% noise. Logistic Regression ranked second at 1.7%, and LightGBM ranked third at 1.9%, which is notable because LightGBM also achieved the best baseline performance on this dataset. Decision Tree remained the weakest model, with a 24.2% relative drop. KNN degraded sharply at 14.8%, while Random Forest showed moderate degradation at 8.5%.

Fig. 2 shows the symmetric degradation curves for the Breast Cancer Wisconsin dataset, and Fig. 3 shows the corresponding curves for the Adult Income dataset. The steeper slopes for Decision Tree and KNN confirm their high sensitivity to random label corruption, while the near-flat curves for SVM and Logistic Regression on the Adult dataset illustrate their strong robustness.

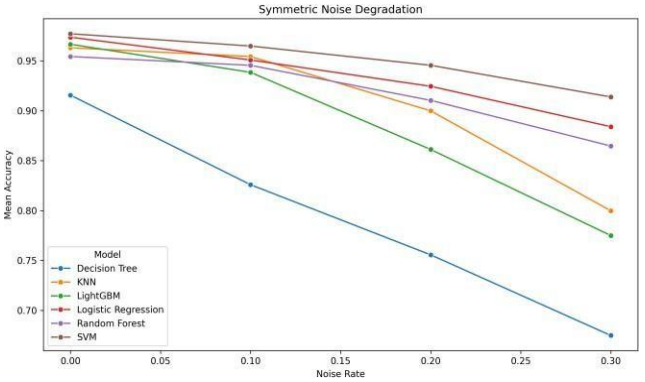

Fig. 2. Accuracy degradation under symmetric noise across noise levels — Breast Cancer Wisconsin dataset

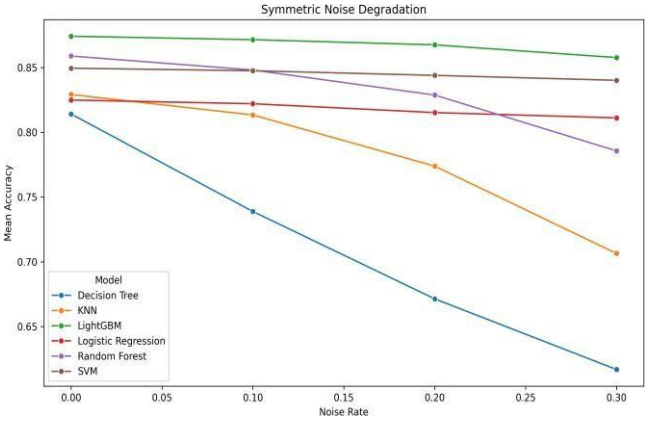

Fig. 3. Accuracy degradation under symmetric noise across noise levels — Adult Income dataset.

Fig. 4 shows the relative accuracy drop under symmetric noise for the Breast Cancer Wisconsin dataset, while Fig. 5 shows the corresponding trend for the Adult Income dataset. Decision Tree exhibits the highest relative drop in both datasets, confirming its strong sensitivity to label corruption. In contrast, SVM shows the smallest relative drop, especially on the Adult

dataset, indicating better robustness. KNN and LightGBM display moderate degradation, with increasing drops at higher noise levels.

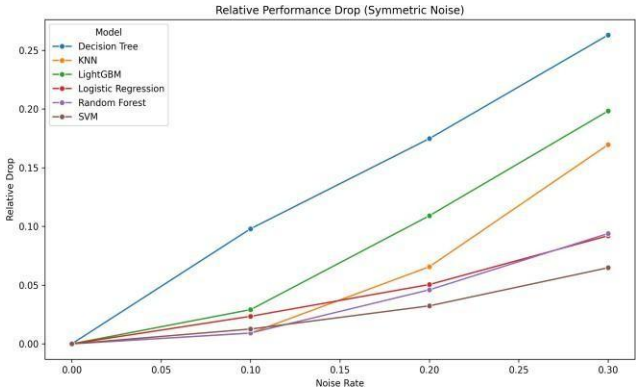

Fig. 4. Relative accuracy drop (%) under symmetric noise — Breast Cancer Wisconsin dataset

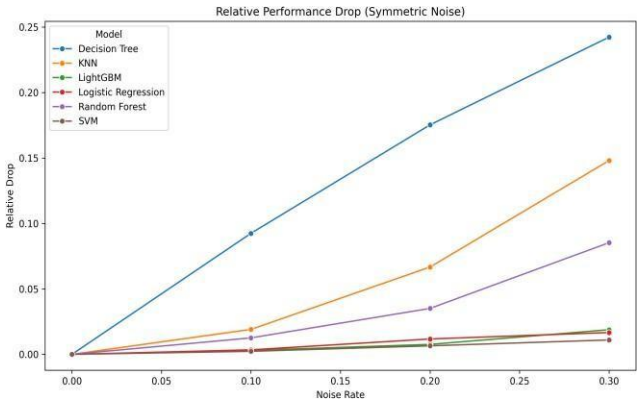

Fig. 5. Relative accuracy drop (%) under symmetric noise — Adult Income dataset.

C. Effect of Asymmetric Label Noise

Asymmetric noise also reduced performance, but its effect was generally smaller than symmetric noise, especially at lower noise rates. The directional nature of asymmetric corruption preserves part of the class structure, which allows models to partially retain discriminative information even under moderate corruption levels.

On the Breast Cancer Wisconsin dataset, SVM again showed the strongest robustness under asymmetric noise, with a total relative drop of only 3.6% at 30% noise. Random Forest was the next most stable with an 8.8% relative drop, followed by Logistic Regression at 10.8%. Decision Tree remained the most affected model with an 18.3% relative drop. KNN and LightGBM occupied the middle range, with relative drops of 13.0% and 12.9%, respectively.

On the Adult dataset, the behaviour changed notably and produced one of the most interesting findings of the study. Under asymmetric noise, KNN achieved the smallest relative accuracy drop of only 2.0% at 30% noise, making it the most robust model in this specific setting. Decision Tree ranked second at 2.2%, and Random Forest third at 2.4%. In sharp contrast, SVM — which had been strongest under symmetric noise — became the most affected model under asymmetric corruption on the Adult dataset, with a 5.5% relative drop. Logistic Regression also showed noticeable decline at 3.9%, especially in F1-score.

This reversal demonstrates that a model's robustness is not a fixed property; it depends on the interaction between model design, data characteristics, and the type of corruption. Asymmetric corruption preserves some local class structure that can still be exploited by neighbourhood-based and tree-based methods on the Adult dataset, whereas globally separating models such as SVM become more affected when one class label is directionally biased.

Fig. 6 and Fig. 7 show the accuracy degradation curves under asymmetric noise for both datasets. The shallower overall slopes compared to symmetric noise confirm that asymmetric corruption is generally less destructive. However, the divergence between models remains clear, especially on Breast Cancer where SVM and Random Forest separate visibly from Decision Tree.

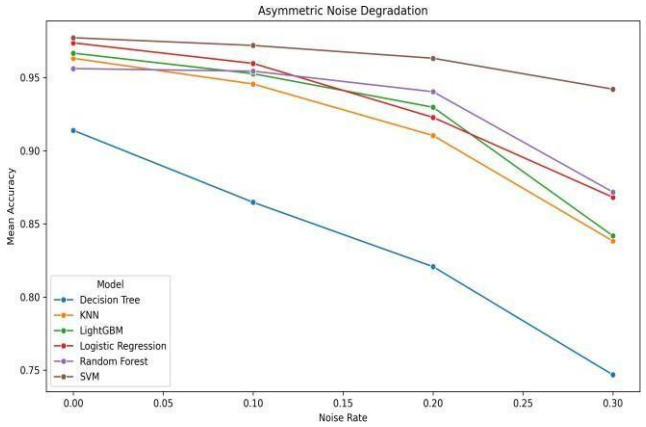

Fig. 6. Accuracy degradation under asymmetric noise across noise levels — Breast Cancer Wisconsin dataset.

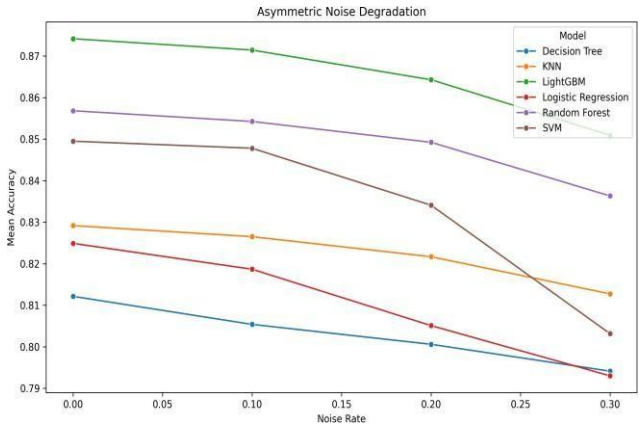

Fig. 7. Accuracy degradation under asymmetric noise across noise levels — Adult Income dataset.

Fig. 8 and Fig. 9 present the relative performance drop under asymmetric noise. The ranking reversal between SVM and KNN on the Adult dataset is clearly visible in Fig. 9, where KNN achieves the smallest drop and SVM the largest among all models.

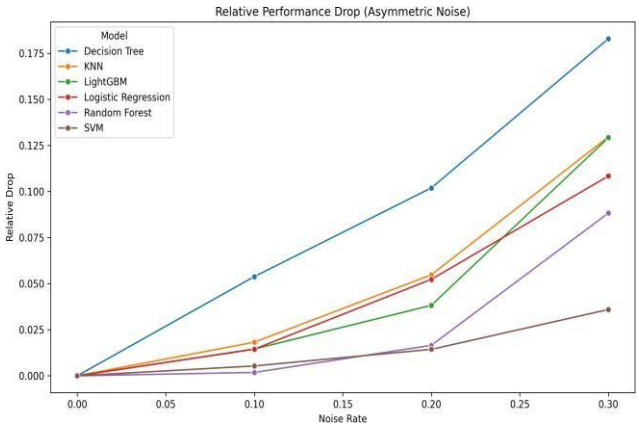

Fig. 8. Relative accuracy drop (%) under asymmetric noise — Breast Cancer Wisconsin dataset.

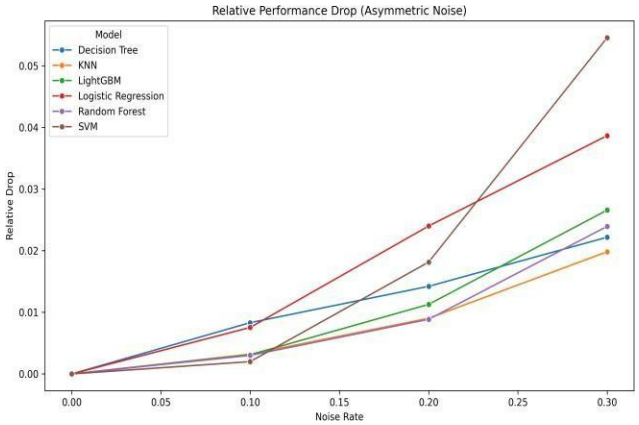

Fig. 9. Relative accuracy drop (%) under asymmetric noise — Adult Income dataset.

D. Comparative Behaviour of Accuracy and F1-Score

An important observation from the experiments is that accuracy alone does not fully describe the impact of label noise. In several cases, the decline in F1-score was stronger than the decline in accuracy, particularly on the Adult dataset under asymmetric noise. This indicates that models may retain acceptable overall accuracy while suffering from a more severe deterioration in their ability to correctly identify both classes.

This effect was especially visible for Logistic Regression and SVM on the Adult dataset, where asymmetric noise caused a pronounced collapse in F1-score compared with the corresponding accuracy drop. On the Breast Cancer Wisconsin dataset, both accuracy and F1-score declined together, but the gap between them remained smaller because the dataset is relatively cleaner and balanced.

Fig. 10 and Fig. 10b present the F1-score degradation under symmetric noise for the Breast Cancer Wisconsin and Adult Income datasets, respectively. Similarly, Fig. 11 and Fig. 11b show the F1-score degradation under asymmetric noise for the two datasets. The asymmetric F1 plots show a more severe divergence between models than the corresponding accuracy plots, especially for Logistic Regression and SVM on the Adult dataset. Our empirical findings confirm that F1-score is a more sensitive and informative robustness metric in the presence of class imbalance, and it should accompany accuracy in noisy- label evaluations.

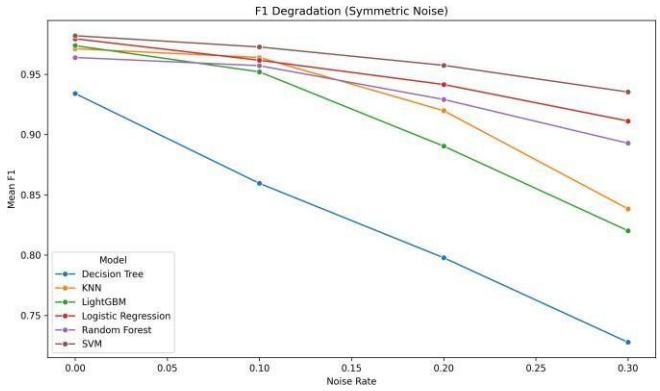

Fig. 10. F1-score degradation under symmetric noise — Breast Cancer Wisconsin dataset.

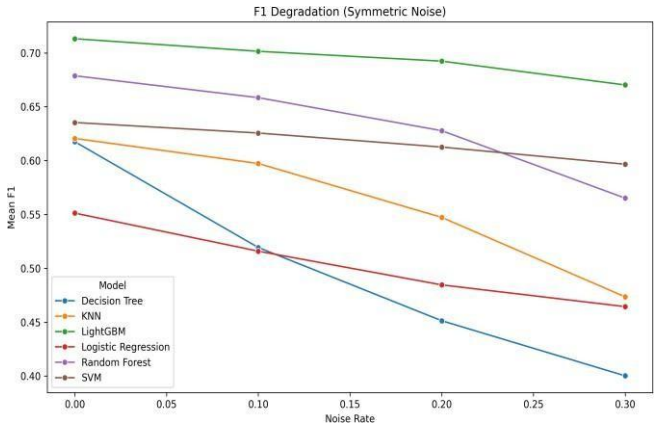

Fig. 10b. F1-score degradation under symmetric noise — Adult Income dataset.

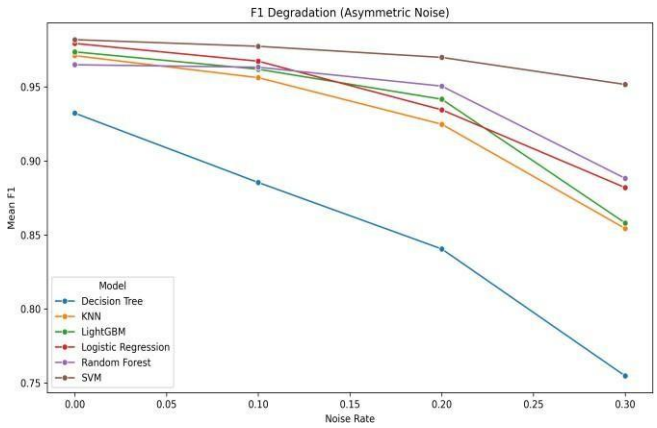

Fig. 11. F1-score degradation under asymmetric noise — Breast Cancer Wisconsin dataset.

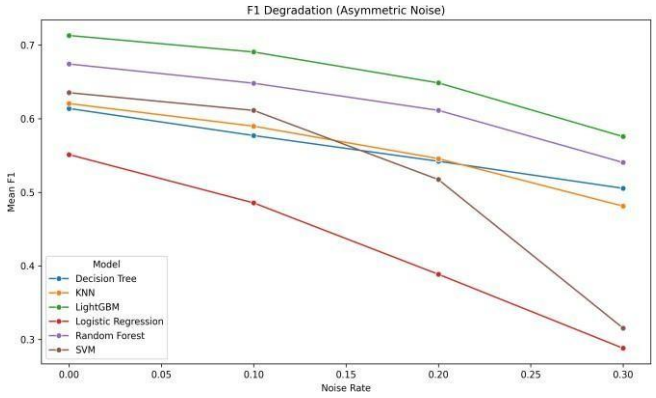

Fig. 11b. F1-score degradation under asymmetric noise — Adult Income dataset.

E. Variance and Stability Across Folds

In addition to mean performance, fold-wise variation provides useful information about robustness. A robust model should not only preserve higher average accuracy, but also remain stable across different train-test partitions of the data.

On the Breast Cancer Wisconsin dataset, standard deviation values were generally higher than on the Adult dataset, which is expected because the smaller dataset size makes each fold more sensitive to local variations in noisy labels. Decision Tree displayed particularly high variability under both noise types, especially at higher symmetric noise levels. KNN and LightGBM also showed increased instability as noise increased. In contrast, SVM and Logistic Regression remained comparatively more stable across most settings.

On the Adult dataset, standard deviations were much smaller overall, reflecting the stabilizing effect of the larger sample size. Still, the same broad pattern remained: Decision Tree and KNN were less stable when noise increased, whereas LightGBM, SVM, and Logistic Regression remained comparatively consistent. Random Forest also showed moderate stability, though generally not at the same level as the most stable models.

Fig. 12 shows mean accuracy with standard deviation bands across noise levels for both symmetric and asymmetric settings. The wider bands around Decision Tree and KNN confirm their greater instability, while the tighter bands around SVM and Logistic Regression reinforce their more consistent behaviour.

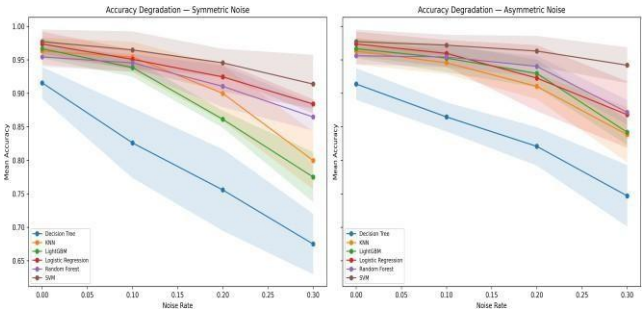

Fig. 12. Mean accuracy with ±1 standard deviation bands across noise levels and models.

F. Cross-Dataset Comparison

Contrasting the two datasets reveals several reliable and consistent patterns. First, baseline performance was much higher on Breast Cancer than on Adult, confirming that robustness must be interpreted relative to dataset difficulty. A model that performs very well on a clean, structured dataset may show better results than on a larger, imbalanced real-world dataset.

Second, Decision Tree was consistently the weekest model in terms of robustness across both datasets and both noise types. This consistency across very different data sources and corruption settings increases confidence in the result and suggests that single-tree learning is naturally sensitive to corrupted labels.

Third, SVM was the most robust model on the Breast Cancer dataset and also the strongest performer under symmetric noise on the Adult dataset, showing that margin-based learning provides strong resistance to random corruption. However, its weaker relative ranking under asymmetric noise on the dataset shows that robustness is noise-type dependent rather than universal.

Fourth, LightGBM and Random Forest were strong all-round performers, especially on the Adult dataset. LightGBM achieved the best clean baseline there and remained among the stronger models across both noise types. However, its advantage over SVM under symmetric noise is notable, as LightGBM showed a higher relative drop on the Adult dataset. Random Forest maintained solid robustness, consistently outperforming Decision Tree. Finally, KNN displayed dataset- and noise-specific behavior: not among the strongest under symmetric noise, but the strongest under asymmetric noise on the Adult dataset.

Table V. Combined comparison of best baseline, best symmetric robustness, best asymmetric robustness, worst model, and strongest F1 collapse over both datasets. Rel. drop = relative accuracy drop at 30% noise.

Criterion	Breast Cancer	Adult Income	Pattern
Best Acc.	SVM (97.7)	LGBM (87.4)	Data-dependent
Best Sym.	SVM (6.5↓)	SVM (1.1↓)	Consistent
Best Asym.	SVM (3.6↓)	KNN (2.0↓)	Noise-dependent
Worst	Decision Tree	Decision Tree	Consistent
F1 Drop	Decision Tree	LR / SVM	Asym. noise

Fig. 13 and 13b present the robustness ranking bar chart for both noise types, summarizing the mean relative drop across all noise levels for each model. Fig. 14 and 14b present the relative drop heatmap across both noise types and all noise levels, enabling simultaneous comparison of all model-noise-level combinations.

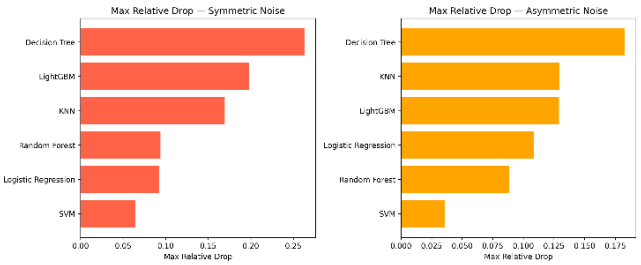

Fig. 13. Robustness ranking of all models by maximum relative accuracy drop at the highest noise level (30%) under symmetric (left) and asymmetric (right) noise - Breast Cancer Wisconsin.

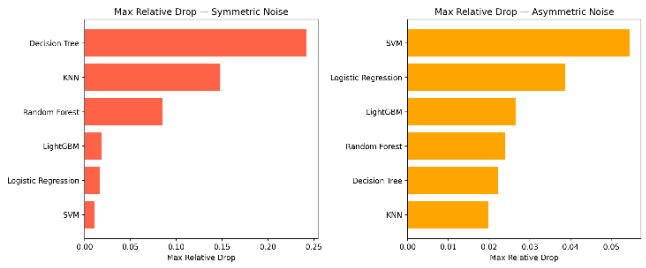

Fig. 13b. Robustness ranking of all models by maximum relative accuracy drop at the highest noise level (30%) under symmetric (left) and asymmetric (right) noise – Adult Income Dataset

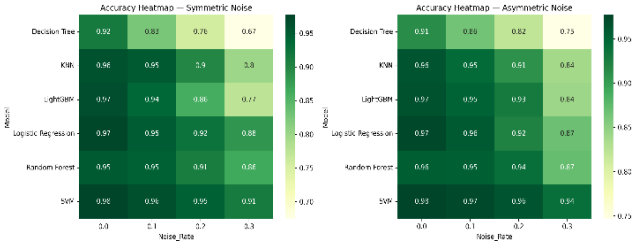

Fig. 14. Accuracy heatmap across all models and noise levels for symmetric (left) and asymmetric (right) noise - Breast Cancer Wisconsin.

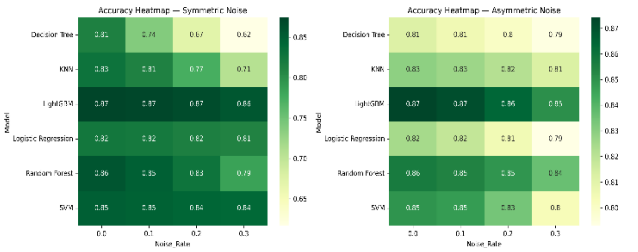

Fig. 14. Accuracy heatmap across all models and noise levels for symmetric (left) and asymmetric (right) noise – Adult Income Dataset

Fig. 15 and 15b show the degradation slope analysis, where steeper bars indicate faster accuracy decline per unit increase in noise rate. Fig. 16 and 16b show the degradation slope analysis, where steeper slopes indicate faster performance decline as noise increases.

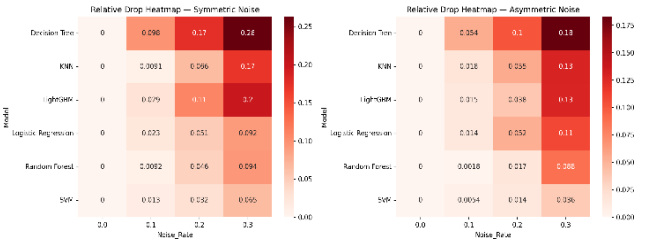

Fig. 15. Relative accuracy drop heatmap for all models and noise levels under both noise types - Breast Cancer Wisconsin.

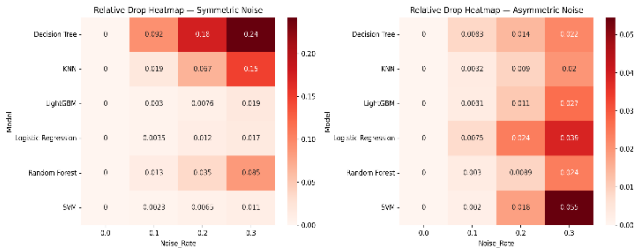

Fig. 15b. Relative accuracy drop heatmap for all models and noise levels under both noise types – Adult Income Dataset.

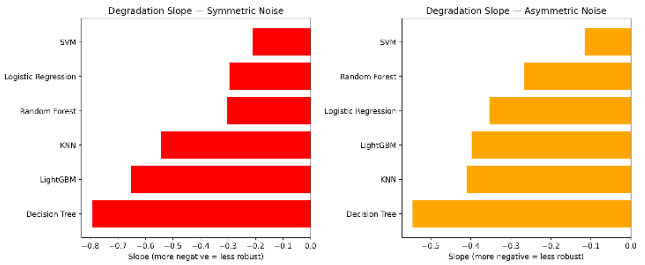

Fig. 16. Degradation slope (linear regression of accuracy on noise rate) for all models under symmetric (left) and asymmetric (right) noise - Breast Cancer Wisconsin.

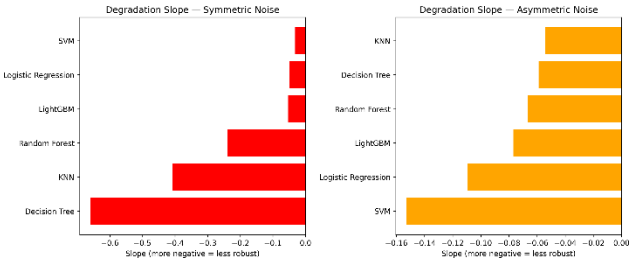

Fig. 16b. Degradation slope (linear regression of accuracy on noise rate) for all models under symmetric (left) and asymmetric (right) noise – Adult Income Dataset

Fig. 17 and Fig. 18 group models by their learning paradigm category and show mean accuracy trends under increasing noise rates, separately for each dataset. On the Breast Cancer Wisconsin dataset, the margin-based category (SVM) consistently shows the flattest degradation curve under both noise types. The tree-based category (Decision Tree) shows the steepest decline. Ensemble and boosting categories fall between these extremes, confirming that aggregation provides partial but not complete protection against label corruption.

On the Adult dataset, the category pattern shifts under asymmetric noise: the instance-based category (KNN) becomes the most stable, overtaking the margin-based category. This cross-dataset divergence reinforces the conclusion that no single learning paradigm is universally most robust; the interaction between model design, data structure, and noise type determines the observed behavior.

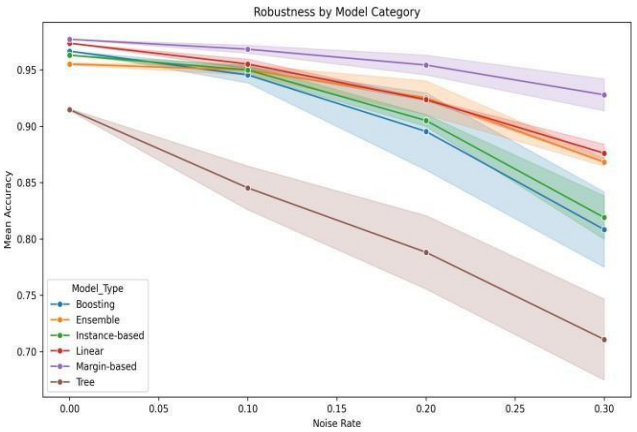

Fig. 17. Mean accuracy by model category across increasing noise rates — Breast Cancer Wisconsin dataset.

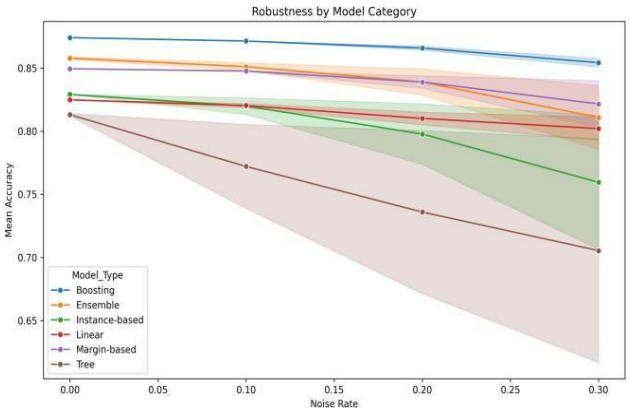

Fig. 18. Mean accuracy by model category across increasing noise rates — Adult Income dataset.

Fig. 19 presents the bar chart comparing model accuracy at the highest noise level (30%) across both noise types and both datasets. This visualization highlights the absolute performance remaining after strong corruption, complementing the relative drop analysis.

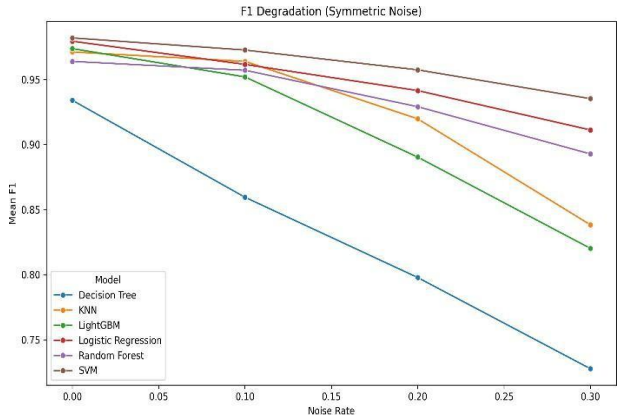

Fig. 19. Mean F1-score degradation under symmetric label noise across increasing noise rates — Breast Cancer Wisconsin dataset.

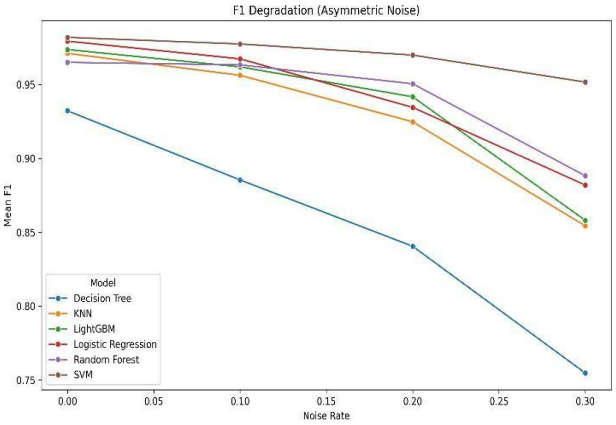

Fig. 20. Mean F1-score degradation under asymmetric label noise across increasing noise rates — Adult Income Dataset

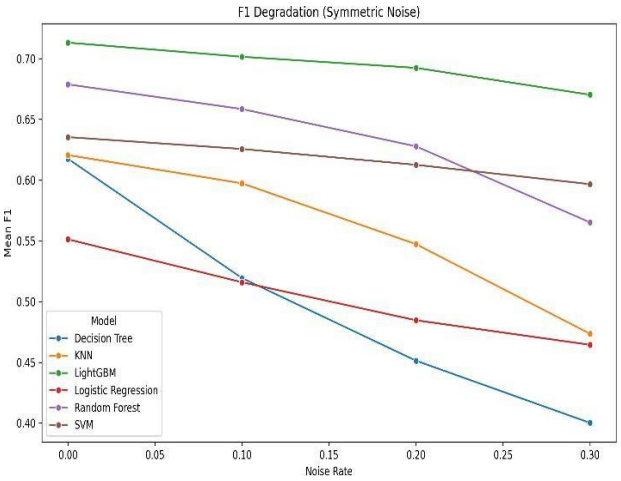

Fig. 21. Mean F1-score degradation under symmetric label noise across increasing noise rates — Adult Income dataset.

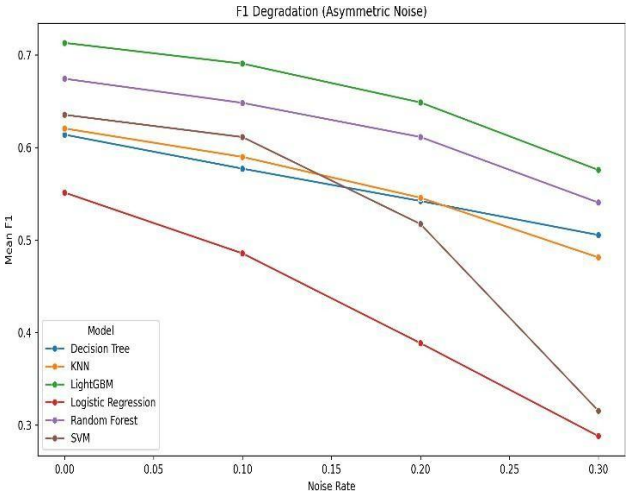

Fig. 22. Mean F1-score degradation under asymmetric label noise across increasing noise rates — Adult Income dataset.

G. Discussion with Respect to Research Questions

RQ1 — How do classical machine learning models behave under label noise? All models degrade as noise increases, but the magnitude of degradation differs substantially across models and datasets. Tree-based and instance-based methods are generally more sensitive, while margin-based and ensemble methods are more robust. The results show that no model is immune to label noise, but some are significantly more tolerant than others.

RQ2 — What is the difference between symmetric and asymmetric noise? Symmetric noise is consistently more damaging than asymmetric noise for most models and on both datasets. This is because symmetric flipping corrupts labels in all directions, eliminating more of the class structure. However, asymmetric corruption can still produce severe decline in some models, particularly in F1-score on the Adult dataset. The reversal of robustness rankings under asymmetric noise on the Adult dataset is a key finding that highlights the interaction between noise type and model design.

RQ3 — Are robustness patterns consistent across datasets? Partially. Some trends are consistent across both datasets, such as the poor robustness of Decision Tree and the strong robustness of SVM under symmetric noise. However, the best-performing model under asymmetric noise changes between datasets (SVM on Breast Cancer, KNN on Adult), means that if we test only one dataset, we can't assure that the results will work the same for other datasets. Cross-dataset evaluation is therefore essential in robustness studies.

H. Key Takeaways from the Results

1. **Clean-data accuracy does not imply robustness.** LightGBM achieved the best baseline on Adult dataset but placed lower than SVM in symmetric robustness.
2. **Symmetric noise is more destructive.** It produced consistently higher relative drops than asymmetric noise across all models and both datasets.

3. **SVM is the most stable under symmetric corruption.** It ranked first on both datasets under symmetric noise, with only 6.5% and 1.1% relative drops at 30% noise.
4. **Decision Tree is consistently the least robust.** It showed the largest relative drops on both datasets under both noise types, with drops exceeding 24%.
5. **Robustness is noise-type- and dataset-dependent.** The KNN reversal on Adult under asymmetric noise shows that robustness rankings are not universal.
6. **F1-score reveals hidden degradation.** Particularly on the imbalanced Adult dataset, F1-score collapsed more severely than accuracy, making it an essential complementary metric.

Taken together, these results demonstrate the need to evaluate model robustness and stability before using it in real-world settings where label quality cannot be guaranteed. A classifier that appears best on clean data may not remain the best once label corruption is introduced, and the choice of evaluation metric matters for accurately measuring that degradation.

I. CONCLUSION AND FUTURE SCOPE

This study checked how six machine learning models work when their training data has labels. It used two sets of data. One about Breast Cancer and the other about Adult Income. The researchers added levels of mistakes to the training data but the test data was always correct. They wanted to see how well the models work when the training data is not perfect.

The results show that when the training data has mistakes the models do not work well.. How much they struggle depends on the model, the data and the type of mistake. Most of the time random mistakes hurt the models more than mistakes that make sense in a way. This was true for how accurate the models were and how well they did at predicting the answers.

It's also important to note that just because a model works with perfect data it doesn't mean it will keep working well with imperfect data. For the Breast Cancer data a model called SVM stayed strong when there were many mistakes. For the Adult data a model called LightGBM did the best with data but SVM did better when there were random mistakes. When the mistakes made sense in a way a model called KNN did the best. This means that no single model is the best in all situations.

The study also found that a model called Decision Tree struggled a lot with mistakes. Its performance dropped a lot. It was not consistent across different tests. On the hand models like Random Forest, LightGBM and SVM were more stable. SVM was especially good at handling mistakes.

The researchers also looked at two ways to measure how well the models work. Accuracy and F1-score. They found that F1-score often shows how much the model struggles clearly and accuracy alone is not enough. This was especially true for the Adult data, where some models seemed to do in terms of accuracy but did much worse in terms of F1-score. This means that using than one way to measure performance gives a better picture of how well the model works with imperfect data.

Table VI summarizes the findings. It shows that the model that does the best with data is not always the best with imperfect data. SVM stayed the strongest with mistakes on both datasets.. With mistakes that make sense in a certain way the best model depended on the data. Decision Tree always did the worst. Also the difference between accuracy and F1-score was more noticeable, on the Adult dataset showing that F1-score reflects the effect of mistakes clearly when the data is not balanced.

Table VI. Combined cross-dataset comparison showing best baseline, best robustness under each noise type, worst model, and F1 collapse pattern. Rel. drop = accuracy drop at 30% noise.

Criterion	Breast Cancer	Adult Income	Pattern
Best Baseline Accuracy	SVM (97.7%)	LightGBM (87.4%)	Dataset-dependent
Best Sym. Robustness	SVM (6.5% rel. drop)	SVM (1.1% rel. drop)	Consistent: SVM
Best Asym. Robustness	SVM (3.6% rel. drop)	KNN (2.0% rel. drop)	Noise-type dependent
Least Robustness	Decision Tree (26.3%)	Decision Tree (24.2%)	Consistent: DT worst
Strongest F1 Collapse	Decision Tree	Logistic Reg. / SVM	Asym. noise on Adult

Overall, the findings answer the research objective clearly: classical machine learning models do not respond equally to noisy labels, and robustness depends on the interaction between learning mechanism, dataset characteristics, and corruption type. Therefore, robustness evaluation should be considered an essential part of model selection when training labels may be imperfect.

A. Future Scope

This work opens up opportunities for further exploration. One way is to add more benchmark datasets so that the results become stronger and can be applied to real-world applications in different domains. Another way is to explore various other noise types like class-dependent, instance dependent or annotation uncertainty. These reflect real data collection scenarios.

It also has the scope to be expanded beyond classical machine learning to deep learning models and noise-aware training methods. Also, there are advanced robustness techniques like cross correction, sample reweighting, confident learning or noise filtering.

Future work may also include statistical significance testing and more theoretical analysis to explain the stability of some models under different corruption settings. Overall, this study provides a solid experimental ground for understanding label-noise robustness in classical machine learning, which can be a basis for more sophisticated application-oriented research on noisy supervised learning.

REFERENCES

[1] W. Wolberg, O. Mangasarian, N. Street, and W. Street. "Breast Cancer Wisconsin (Diagnostic)," UCI Machine Learning Repository, 1993. [Online]. Available: https://doi.org/10.24432/C5DW2B.

[2] B. Becker and R. Kohavi. "Adult," UCI Machine Learning Repository, 1996. [Online]. Available: https://doi.org/10.24432/C5XW20.

[3] C. M. Bishop, Pattern Recognition and Machine Learning. New York, NY, USA: Springer, 2006.

[4] L. Breiman, "Random forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[5] V. N. Vapnik, Statistical Learning Theory. New York, NY, USA: Wiley, 1998.

[6] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," Annals of Statistics, vol. 29, no. 5, pp. 1189–1232, 2001.

[7] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in Proc. NeurIPS, 2017, pp. 3146–3154.

[8] D. Angluin and P. Laird, "Learning from noisy examples," Machine Learning, vol. 2, no. 4, pp. 343–370, 1988.

[9] B. Frenay and M. Verleysen, "Classification in the presence of label noise: A survey," IEEE Trans. Neural Netw. Learn. Syst., vol. 25, no. 5, pp. 845–869, 2014.

[10] X. Zhu and X. Wu, "Class noise vs. attribute noise: A quantitative study," Artificial Intelligence Review, vol. 22, pp. 177–210, 2004.

[11] S. Rolnick et al., "Deep learning is robust to massive label noise," arXiv preprint arXiv:1705.10694, 2017.

[12] J. Patrini et al., "Making deep neural networks robust to label noise: A loss correction approach," in Proc. CVPR, 2017, pp. 1944–1952.

[13] C. Northcutt, T. Wu, and I. Chuang, "Confident learning: Estimating uncertainty in dataset labels," J. Artificial Intelligence Research, vol. 70, pp. 1373–1411, 2021.

[14] N. Natarajan et al., "Learning with noisy labels," in Proc. NeurIPS, 2013, pp. 1196–1204.

[15] C. Cortes and V. Vapnik, "Support-vector networks," Machine Learning, vol. 20, no. 3, pp. 273–297, 1995.

[16] E. Fix and J. L. Hodges, "Discriminatory analysis: Nonparametric discrimination: Consistency properties," USAF School of Aviation Medicine, Project 21-49-004, Report No. 4, 1951.

[17] L. Rokach, "Decision trees: A survey," Data Mining and Knowledge Discovery, vol. 14, no. 1, pp. 1–26, 2006.